



Nutritional interventions in patients with graft-versus-host disease

David Seguy^{a,b,c} and Thomas Hueso^{d,e}

Purpose of review

This review aims to highlight the benefits of nutrition before and during graft-versus-host disease (GvHD) and the promising precision medicine approach that should be offered to prevent and mitigate GvHD.

Recent findings

The intestinal damage induced by preconditioning/conditioning chemotherapies is the main trigger of GvHD. Impaired nutritional status and decreased plasma citrulline level, which is the most sensitive biomarker of intestinal barrier health, predict the occurrence of acute GvHD after allogeneic hematopoietic cell transplantation (allo-HCT). Optimal oral and/or enteral nutrition and a lack of vitamin D deficiency limit this intestinal damage. As intestinal dysbiosis plays an important role in GvHD, probiotics and prebiotics supplementation could be a promising therapy. Diverting enterostomy combined with parenteral nutrition saves the lives of patients with severe steroid-refractory gastrointestinal GvHD.

Summary

Regardless of age, healthy nutritional status and a healthy gut barrier are protective factors against GvHD in patients undergoing allo-HCT, and above all, these are closely dependent on adequate oral and/or enteral intake. Therefore, maintaining gut barrier integrity through adequate oral nutrition before allo-SCT and early first-line enteral nutrition after allo-HCT are of critical importance, not forgetting vitamin D supplementation. In the future, probiotics and prebiotics are expected to play a growing role for replenishing the commensal microbiota given the impact of gut dysbiosis on GvHD. Parenteral nutrition remains the only nutritional support that can be used in the event of severe gastrointestinal GvHD.

Keywords

allogeneic hematopoietic cell transplantation, chemotherapy, citrulline, enteral nutrition, intestinal barrier

INTRODUCTION

Allogeneic hematopoietic cell transplantation (allo-HCT) remains the standard of care for patients with hematological malignancies and, more specifically, for patients with acute leukemia eligible for intensive treatment. This procedure consists of the administration of conditioning chemotherapy followed by a graft infusion from a healthy human leucocyte antigen (HLA)-matching donor. It combines the cytotoxic effect of chemotherapy, allowing graft expansion and the proliferation of alloreactive T lymphocyte cells (T-cells) against leukemia called the graft-versus-leukemia (GvL) effect. The goal is to maximize the GvL effect to control the underlying disease while restraining the graft-versus-host (GvH) effect, which constitutes the other side of the same coin. Acute graft-versus-host disease (GvHD) remains one of the most serious complications within the first 3 months after the graft infusion. Almost 40% of patients develop GvHD, and 5–15% of patients experience severe

GvHD, which is responsible for high mortality rates [1]. The murine model by Ferrara *et al.* [2] describes this physiopathology. Thus, conditioning chemotherapy is responsible for gut damage, which leads to the following (Fig. 1): release of proinflammatory cytokines and bacterial intestinal translocation; activation of host antigen-presenting cells and donor-derived T cells; and production of further cytokines and tissue apoptosis that exacerbate alloractivity and cytokine storm [2].

^aUniversity of Lille, U1286 - Infinite - Institute for Translational Research in Inflammation, ^bInserm, U1286, ^cCHU Lille, Nutrition, Lille, ^dClinical Hematology Department, Avicenne Hospital, Assistance Publique-Hôpitaux de Paris (AP-HP) and ^eSorbonne University Paris-Nord, Bobigny, France

Correspondence to Pr. David Seguy, Faculté de Médecine - Pôle Recherche, 1 place de Verdun, 59045 Lille, France.
E-mail: david.seguy@univ-lille.fr

Curr Opin Clin Nutr Metab Care 2023, 26:455–462

DOI:10.1097/MCO.0000000000000963

KEY POINTS

- Intensive chemotherapy induces intestinal damage and gut dysbiosis, which play a central role in triggering GvHD.
- Low plasma citrulline level is the most sensitive marker of intestinal epithelial barrier damage, predicting GvHD in the pretransplant period and correlating with its severity in the posttransplant period.
- Strengthening the intestinal barrier through oral and/or enteral nutrition is of crucial importance to limiting GvHD and its related morbidity and mortality.
- All patients should be supplemented with vitamin D before and after undergoing allo-HCT.
- Proximal diverting enterostomy combined with parenteral nutrition is the lifeline of steroid-refractory gastrointestinal GvHD.

Severe GvHD is strongly associated with gut damage that leads to intestinal failure, malnutrition, escalation of immunosuppressive therapy that promotes sepsis, and death in cases of uncontrolled disease. This work looks beyond the immunosuppressive approach to limit the donor's T cell alloreactivity. It asks the following questions: what are the benefits of nutrition before and during GvHD? What nutritional approaches should doctors offer to prevent and mitigate GvHD? This review focuses on acute GvHD. It will not address chronic GvHD as currently, nutritional support is known to help fight against malnutrition but does not influence the progression of the disease.

BEFORE THE STORM

Role of nutritional status

In the months leading up to the allo-HCT procedure, induction and consolidation chemotherapies performed to control the underlying disease induces side effects (diarrhea, nausea, mucositis, and infections) that potentiate the disease's catabolic and anorectic effects, harming one's nutritional status. Malnutrition before and during allo-HCT is a recognized risk factor that influences short-term clinical outcome, survival, and GvHD regardless of age [3,4⁵,5,6]. A study evaluated a composite anthropometric score in a pediatric population using a ROC curve. The sum score, defined as the addition of the Z-score of weight for age and the Z-score of height for age at the time of allo-HCT, was predictive of severe GvHD with a sensitivity of 0.929 and a 1-specificity of 0.873 [5].

Serum albumin remains the best biological marker indicating the intensity of the inflammatory response and protein catabolism. In multivariate analysis, a serum albumin level less than 30 g/dL before allo-HCT remains the strongest predictor of 100-day nonrelapse mortality (NRM) in adults [7]. In a cohort of 970 adults with acute myelogenous leukemia undergoing allo-HCT, decreasing pretransplant albumin levels were also associated with NRM [hazard ratio 0.36; 95% confidence interval (95% CI) 0.25–0.53; $P < 0.001$] [8]. In pediatric patients, albumin level less than 30 g/dL, weight loss at least 10%, and BMI less than 5th percentile at the beginning of allo-HCT predicted a significant risk of developing severe (grade III-IV) GvHD [6].

Role of gut health status

Preconditioning chemotherapy is highly toxic to intestinal epithelium turnover, resulting in gut damage and alteration of the microbiota even before the allo-HCT procedure. Of the measurable surrogate markers of intestinal health, studies have mentioned the plasma citrulline level the most, with four relevant retrospective studies discussing it within the last 10 years [9¹⁰]. This nonprotein amino acid is exclusively produced by enterocytes (which are located in the small intestine) from glutamine contained in proteins taken orally and/or enterally. It is consecutively released into the bloodstream, taken up by the kidneys, and used for arginine biosynthesis. It was initially used to assess enterocyte functional mass in small bowel syndrome to estimate the long-term parenteral nutrition dependency. Later, hematologists began using citrulline to monitor chemotherapy-induced mucositis [10]. Since then, our team has suggested that intestinal damage induced by induction and consolidation chemotherapies could bring down pretransplant citrulline levels and enhance bacterial translocation priming macrophage reactivity, which both trigger GvHD in the posttransplant period [11]. To support our hypothesis, we assessed plasma citrulline levels in a larger cohort of patients undergoing allo-HCT and found that a citrulline level 26 $\mu\text{mol/l}$ or less 1 month before the initiation of conditioning chemotherapy is an independent factor of gastrointestinal GvHD development and NRM [12]. To better understand how prepreconditioning citrulline levels could impact GvHD onset, our team evaluated the effects of standard chemotherapy used to treat acute leukemia in humans and a murine model. Along with decreasing citrulline, chemotherapies (mainly cytarabine and anthracycline) induce a decrease in short-chain fatty acids (SCFA) and a loss of microbiota diversity (alpha and beta diversity),

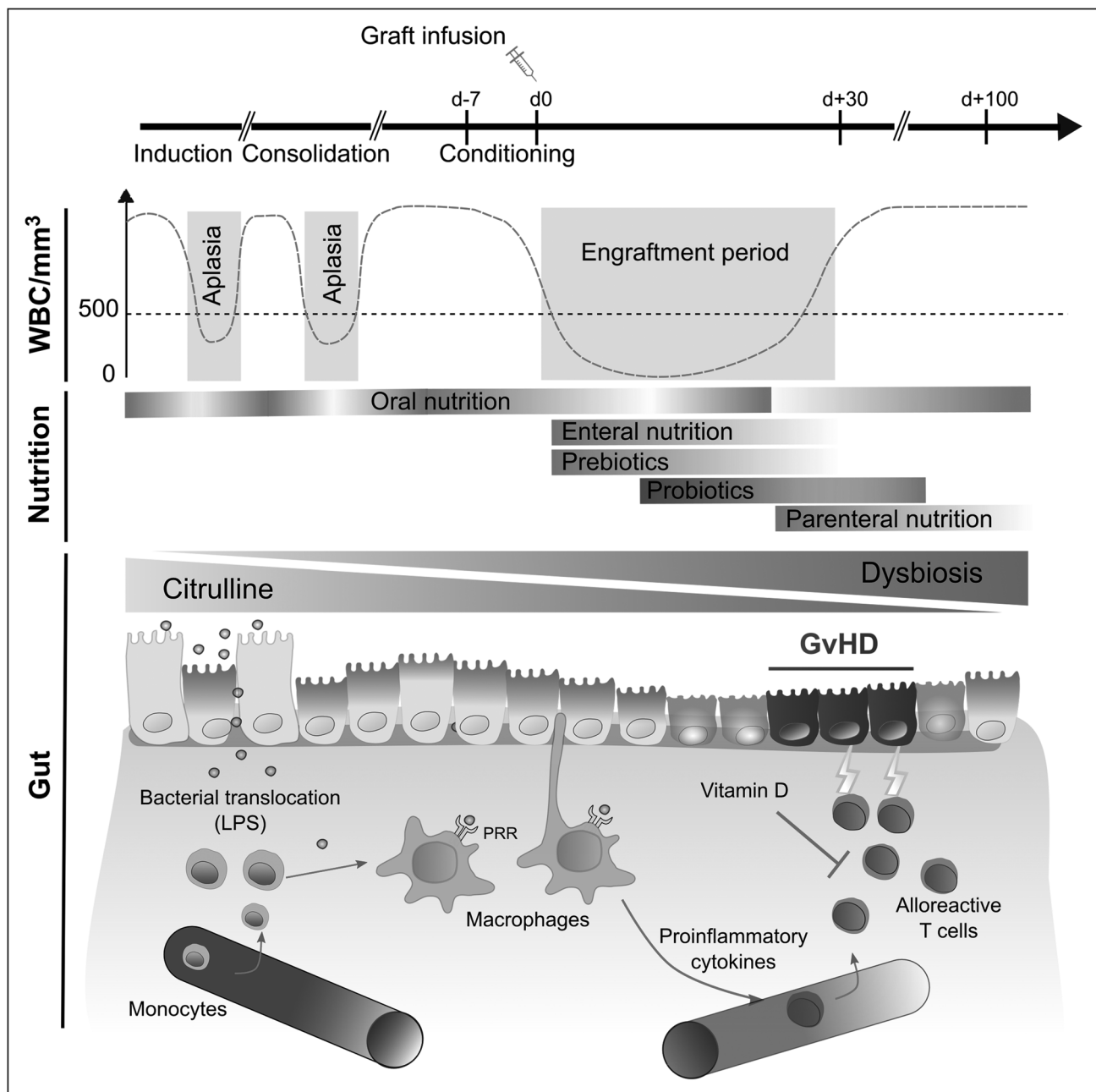


FIGURE 1. Acute graft versus host disease pathophysiology and its nutritional management. Induction and consolidation chemotherapies are administered to obtain remission of hematological disease. Conditioning chemotherapy is followed by graft infusion from a healthy donor at d0. A sequential chemotherapy regimen is responsible for the aplasia period and successive gut barrier impairment is responsible for further dysbiosis and bacterial translocation. Before the allogeneic hematopoietic cell transplantation procedure, LPS (lipopolysaccharide) may prime macrophage reactivity responsible for proinflammatory cytokine release. During the posttransplant period, alloreactive T-cells from donor are stimulated, inducing the production of further cytokines and tissue apoptosis that exacerbate alloreactivity and cytokine storm. GvHD, acute graft versus host disease; PRR, pattern recognition receptors; WBC, white blood cells.

shifting to more aerotolerant bacteria and less anaerobic bacteria [13]. The dysbiosis and the decrease in citrulline may persist until the allo-HCT procedure, resulting in increased bacterial translocation, host antigen-presenting cell activation, and severe GvHD [11,12,14[¶]].

Apart from enterocyte mass, surrogate markers of intestinal barrier disruption (stool zonulin a component of tight junction, calprotectin, and β -defensin 2) were assessed, but failed to correlate with the lactulose mannitol ratio, which was elevated in more than half of patients before the conditioning

chemotherapy [15]. Therefore, regenerating islet-derived 3 α (Reg3 α) serum levels at day 0 of allo-HCT correlates with gut dysbiosis and predicts 1-year transplant-related mortality [16]. Reg3 α is a promising biomarker that rises in the bloodstream, as GvHD progressively destroys Paneth cells and damages the intestinal epithelial barrier.

An increasing number of studies have shown that chemotherapy, reduction of oral intake, and broad-spectrum antibiotics play important roles in the pathophysiology of GvHD through the loss of gut microbiota diversity and the domination of a single bacterial taxon. The composition of oral and gut microbiota influences GvHD occurrence and severity [17,18]. One study revealed the role of *Actinomyces* and *Prevotella melaninogenica* in the oral cavity and of *Lactobacillus* in the gut of children before the allo-HCT procedure in predicting severe (grade III-IV) GvHD. The study suggested that some bacteria might have probiotic properties, promoting immune homeostasis and thereby exerting a protective effect against other bacteria. The authors suggested that *Lactobacillus* expansion may limit enterococcal dominance and promote T-cell reconstitution [19]. Furthermore, in adults, a higher relative abundance of *Streptococcus* and *Corynebacterium* in the supragingival dental biofilm before conditioning was associated with a higher risk of GvHD; conversely, a higher abundance of *Veillonella* predicted a lower risk [20]. Characterization of gut microbiota performed on 1362 patients undergoing allo-HCT showed that the risk of GvHD-related death was lower among patients with higher diversity than among those with lower diversity (hazard ratio 0.49 95% CI 0.26–0.90). This study associated *Enterococcus* dominance with a higher risk of GvHD. Further studies are necessary to better define the bacteria that contribute to gut health before allo-HCT and those that trigger GvHD. In the future, probiotics and prebiotics are expected to play an increasing role in combating GvHD considering the impact of intestinal dysbiosis on the disease.

Feeding the gut

Healthy nutritional status and a healthy gut barrier are protective factors against GvHD, and they are dependent on adequate oral and/or enteral intake. Therefore, strengthening the gut barrier by maintaining optimal oral nutrition before and initiating early first-line enteral nutrition after allo-HCT is critically important.

Oral feeding

In the pretransplant period, a study showed that oral intake inadequacy (that promotes malnutrition)

was related to more posttransplant GvHD ($P < 0.05$), suggesting that maintaining optimal intestinal feeding plays a protective role [21]. This was consistent with a retrospective study comparing a low-bacterial neutropenic diet to a general hospital diet following auto/allo-HCT. The study found that the rate of infections and grade II-IV GI-GvHD at 60 days posttransplant were higher in patients on the neutropenic diet (because of the dietary restriction it involved) compared with those on a general diet [22]. A meta-analysis confirmed that a neutropenic diet cannot reduce the risk of bloodstream infection in patients treated for acute leukemia [23]. Similarly, the longer one fasts after completing conditioning chemotherapy, the higher the frequency of GvHD. Indeed, in humans, glutamine (contained in proteins) provided orally/enterally constitutes the main energy source necessary to maintain enterocyte turnover every 4 days, and then epithelium trophicity is reflected by plasma citrulline levels. Thus, the first form of nutritional support provided to patients should be dietary counseling and food fortification, without forcing, to help manage symptoms. Diet practice in patients undergoing allo-HCT in various expert centers is very heterogeneous, but most centers provide a low-microbial diet during the neutropenic phase [24]. Currently, the Society of Parenteral and Enteral Nutrition (ESPEN) indicates that there is insufficient consistent clinical data to recommend a low-microbial diet for patients more than 30 days after allo-HCT [3]. Furthermore, ESPEN recommends patients maintain physical activity and ensure adequate nutritional intake. Doctors may advice the additional use of oral nutritional supplements when the diet is insufficient to meet nutritional goals, but they have no reported effect on GvHD [4[¶]]. Although oral glutamine supplementation was shown to reduce the incidence of GvHD, no studies have confirmed this effect in the last 15 years.

Probiotics and prebiotics

There is growing evidence that diet may modulate gut microbiota and offer a promising approach to mitigate GvHD.

The administration of probiotics in hematological patients with altered gut permeability remains debated and controversial due to the risk of bacterial translocation and bloodstream infection. A few studies have demonstrated a possible reduction of GvHD with oral feeding using *Lactobacillus*, but the benefit of probiotic supplementation during allo-HCT still needs to be determined [25]. Furthermore, the timing of probiotic administration appears critical. It is certainly deleterious to administer it immediately after the conditioning chemotherapy when

the risk of intestinal bacterial translocation is greatest. Conversely, administering probiotics could be remotely beneficial for replenishing the commensal microbiota to counteract intestinal dysbiosis. Promising results on the safety and efficacy of fecal microbiota transplantation in the treatment of GvHD sustain this hypothesis [26].

Thus, it appears that the prebiotic is safer. Prebiotic galacto-oligosaccharides demonstrated a reduction of GvHD when starting *ad libitum* 7 days before and continuing until day +100 after allo-HCT in a murine model. The prebiotic was associated with an increase in SCFA and butyrate production [27]. However, a preclinical study also demonstrated that donor T cells deficient in the SCFA receptor induce decreased GvHD [28]. Other prebiotic mixtures and resistant starch were used in humans undergoing allo-HCT, decreasing the severity of oral mucositis and grade II-IV GvHD [25]. Conversely, data obtained from murine models and humans demonstrate a negative effect of lactose on *Enterococcus* dominance, which enhances GvHD, suggesting that lactose intake should be limited [29].

Vitamins

Several recent publications have reported on the potential role of fat-soluble vitamins in mitigating or enhancing GvHD. Retinoic acid and vitamin A seem to increase intestinal permeability, with an increase in serum lipopolysaccharide (LPS) levels and the expansion of opportunistic pathogens such as *Escherichia-Shigella* and *Clostridium perfringens* in an allo-HCT murine model, producing a higher incidence of GvHD. However, the underlying mechanism is not clear and results are inconsistent with previous clinical studies [30].

Vitamin D's role in GvHD has been widely documented in the last two years and may play a key role in GvHD pathogenesis: a clinical study demonstrated that a low vitamin D3 level, defined as a level below 20 ng/ml before allo-HCT, was associated with a higher incidence of GvHD in a pediatric cohort suffering from thalassemia major undergoing allo-HCT ($N=100$) [31[■]]. Another study found that a pediatric cohort treated with ultra-high vitamin D supplementation had a lower GvHD incidence [32]. The study by Macedo *et al.* [33] accurately described the immunomodulatory role of vitamin D, reporting that high levels may down-regulate T cell proliferation via epigenetic gene modulation (EZH2 and ID3), whereas vitamin D deficiency may induce high production of proinflammatory cytokines such as tumor necrosis factor α (TNF- α), interleukin 2 (IL-2), IL-4, or interferon γ (INF- γ). Furthermore, vitamin D could have a synergistic action with antithymocyte globulin (ATG)

serum administered to prevent GvHD: *in vitro*, ATG increases gene expression of CYP27B1, the enzyme responsible for the conversion of 25-hydroxyvitamin D3 into 1,25-dihydroxy vitamin D3, known to induce tolerogenic dendritic cells and T regulatory cells [34]. Apart from the immunomodulatory effect of vitamin D, the vitamin D receptor (VDR) plays a role in intestinal homeostasis. VDR is expressed by LGR5+ intestinal stem cells, which modulate epithelial regeneration. Matos *et al.* [35] have assessed intestinal biopsies in patients with severe GI-GvHD and found that there is a significant downregulation of VDR gene expression in these patients compared to those with mild or no GI-GvHD. The reduction of VDR expression was already detectable at the GI-GvHD onset compared to GvHD-free patients and correlated with antimicrobial peptide gene expression [35]. To summarize, all patients should take vitamin D supplements before and after undergoing allo-HCT.

Instead, few data are available concerning vitamin E. We record a recent retrospective study that did not confirm the association between low pretransplant vitamin E and increased risk of GvHD [36]. Thus, there are no conclusive data on its effects.

Enteral or parenteral nutrition?

Feeding the gut and strengthening the intestinal epithelial barrier before and during the allo-HCT procedure are major challenges. Consequently, as recommended by ESPEN with a strong consensus, oral and enteral nutrition should be preferred to parenteral nutrition [3]. However, underlying disease, pretransplant conditioning, posttransplant conditions, and complications, especially those involving the digestive tract, contribute to limiting oral feeding. Beyond the significant positive effect of enteral nutrition on early survival, neutrophil, and platelet engraftment, we showed that compared with parenteral nutrition, an upfront overnight enteral nutrition (initiated at day 1 after allo-HCT and combined with usual oral feeding during the day) decreases GvHD in both children and adults [37,38]. In our prospective cohort study of 121 adult patients, enteral nutrition was a protective factor against severe (grade III-IV) GvHD development (hazard ratio 0.19, 95% CI, 0.05–0.72, $P=0.01$). Similarly, we observed a decrease in grade II-IV GvHD (37 versus 54%, $P=0.0127$) in a retrospective pediatric study wherein 97 children receiving early enteral nutrition were closely matched with patients from three other centers receiving parenteral nutrition instead [37]. The fact that enteral nutrition particularly reduced gastrointestinal and hepatic GvHD suggests that proinflammatory cytokines and translocated bacterial products are imported

from the damaged intestine to the liver via the portal vein. These results have since been integrated and confirmed in a meta-analysis reporting lower incidence rates of all grades of GvHD (relative risk (RR) 0.69, 95% CI 0.56–0.86, $P=0.0007$), grade III–IV GvHD (RR 0.44, 95% CI 0.30–0.64, $P<0.0001$), and GI-GvHD (RR 0.44, 95% CI 0.30–0.66, $P<0.0001$) in the EN group compared with the PN group [39^{***}].

As it is difficult and ethically questionable to perform small bowel/colon biopsies in such patients, the beneficial effects of enteral nutrition can be inferred from the knowledge of enteral nutrition properties on gut physiology and GvHD pathophysiology. Firstly, early enteral nutrition allows to compensate for a reduction (because of anorexia) or interruption (because of painful mucositis) of oral intake after allo-HCT. Unlike parenteral nutrition, enteral nutrition is known to stimulate epithelial trophicity, limiting intestinal damage subsequent to the conditioning regimen and thus the production of self-antigens, proinflammatory cytokines, and intestinal bacterial translocation, which exacerbate the alloreactivity of the donor T cell. Enteral nutrition may maintain homeostasis of the intestinal microbiome, limiting *Enterococcus* proliferation [40]. Furthermore, unlike food, enteral nutrition formulas are gluten-free, lactose-free, and germ-free. The use of hyperproteic (up to 100 g/l) and hypercaloric formulas (up to 2 kcal/ml) allows practitioners to improve patients' tolerance by reducing the volume administered and providing more than 10 g of glutamine/l. Moreover, such formulas can contain fibers that produce SCFA in the colon and anti-inflammatory omega-3 fatty acids.

Typically, polymeric enteral nutrition formulas are prescribed, but a study with a formula containing TGF- β 2, which can be administered orally or enterally, also demonstrated the benefit of limiting malnutrition and GvHD using the oral route [41]. Enteral nutrition is most often infused *via* a nasogastric tube. However, given the guidelines for the use of enteral nutrition in the first line, the foreseeable duration of enteral nutrition, and the risk of displacement of the nasogastric tube, among other things, an increasing number of pediatricians recommend the use of gastrostomy in children undergoing allo-HCT [42]. A cohort study confirmed that percutaneous endoscopic gastrostomy placement was associated with positive transplant outcomes in such patients, but did not observe any influence on the development of GvHD [43].

Conversely, parenteral nutrition is unable to fight against epithelial atrophy following intestinal fasting. Parenteral nutrition is also responsible for more catheter-related infections, metabolic

disorders (hyperglycemia, hypertriglyceridemia, hepatic disorders), thrombopenia, and complications linked to allo-HCT procedures such as sinusoidal obstruction syndrome [4[■],37,38,39^{***},40,44].

DURING THE STORM

Skin and liver graft-versus-host disease

Acute skin or liver GvHD, even when severe, does not require the use of parenteral nutrition, especially in cases of hepatic disorders. Hyperproteic oral and enteral nutrition should be continued to enhance epithelial trophicity and avoid further intestinal damage. The current practice appears more heterogeneous in the case of GI-GvHD, as concluded by a survey performed in 72 German transplant centers. Most practices applied a special diet: 54% supplied a lactose-free diet, 43% a fat-free/reduced diet, and 18% a gluten-free diet. Only 14% of centers stopped oral intake in more severe cases [24].

Gastrointestinal graft-versus-host disease

The prognosis of GvHD is closely linked to gut damage, which is responsible for severe malnutrition and dehydration with a high rate of mortality. Treatments are based on immunosuppressive drugs such as steroids to limit the proliferation of alloreactive T cells and further tissue apoptosis. ESPEN suggests the use of parenteral nutrition to avoid further malnutrition in the case of GI-GvHD, but few data are available [3]. GI-GvHD with severe diarrhea makes parenteral nutrition unavoidable, but this does not prevent the administration of small amounts of EN for trophic purposes. The persistence, despite second line of immunosuppression, of grade III–IV steroid-refractory GI-GvHD can justify a proximal diverting enterostomy performed 100–150 cm from the Treitz angle, excluding the distal ileum segment, where Peyer's patches are the most abundant and GvHD the most active. In our experience, these patients suffering from deep intestinal failure (figures by a plasma citrulline level $< 5 \mu\text{mol/l}$) are at a high risk of death from malnutrition and sepsis. Compared with usual immunosuppressive treatment, we have shown that this diverting enterostomy increased overall survival at 1-year and 2-year [54% ($n=7/13$) versus 5% ($n=2/38$), $P=0.0004$ and 31% ($n=4/13$) versus 2.5% ($n=1/38$), $P=0.0015$] [45^{***}]. After surgery, we encouraged oral feeding in our patients, but we also systematically initiated personalized parenteral nutrition via central venous access for a foreseeable duration of 6–9 months until the restoration of their digestive continuity.

CONCLUSION

Currently, the prevention and treatment of GvHD are mainly based on the administration of immunosuppressive drugs aimed at inhibiting the alloreactivity of the donor's T cells. However, these treatments dramatically aggravate the morbidity and mortality associated with the occurrence of severe sepsis. Strengthening the gut barrier before and during the allo-HCT through optimal oral and/or enteral nutrition constitutes a smart alternative to keep GvHD under control. It may be of interest to develop new oral/enteral formulas, which beyond their nutritional properties, will specifically reinforce each element constituting the intestinal barrier: commensal microbiota, mucus, enterocytes and colonocytes, and gut-associated lymphoid tissue.

Acknowledgements

None.

Financial support and sponsorship

None.

Conflicts of interest

There are no conflicts of interest.

REFERENCES AND RECOMMENDED READING

Papers of particular interest, published within the annual period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Cahn JY, Klein JP, Lee SJ, *et al.* Prospective evaluation of 2 acute graft-versus-host (GVHD) grading systems: a joint Société Française de Greffe de Moëlle et Thérapie Cellulaire (SFGM-TC), Dana Farber Cancer Institute (DFCI), and International Bone Marrow Transplant Registry (IBMTR) prospective study. *Blood* 2005; 106:1495–1500.
2. Ferrara JL, Levine JE, Reddy P, Holler E. Graft-versus-host disease. *Lancet* 2009; 373:1550–1561.
3. Muscaritoli M, Arends J, Bachmann P, *et al.* ESPEN practical guideline: clinical nutrition in cancer. *Clin Nutr* 2021; 40:2898–2913.
4. Muratore E, Leardini D, Baccelli F, *et al.* The emerging role of nutritional support in the supportive care of pediatric patients undergoing hematopoietic stem cell transplantation. *Front Nutr* 2023; 10:1075778.
5. Dos Santos Nunes Pereira AC, Chahin BM, Tarzia A, Vilela RM. Nutritional status and prognosis in children with immunodeficiencies undergoing hematopoietic stem cell transplantation. *Clin Nutr* 2022; 52:1–11.
6. Kerby EH, Li Y, Getz KD, *et al.* Nutritional risk factors predict severe acute graft-versus-host disease and early mortality in pediatric allogeneic hematopoietic stem cell transplantation. *Pediatr Blood Cancer* 2018; 65:e26853.
7. Schaffrath J, Diederichs T, Unverzagt S, *et al.* Correlation of nutrition-associated parameters with nonrelapse mortality in allogeneic hematopoietic stem cell transplantation. *Ann Hematol* 2022; 101:681–691.
8. Orvain C, Byelykh M, Othus M, *et al.* Relationship between pretransplantation nutritional status and outcome in adults with acute myelogenous leukemia undergoing allogeneic hematopoietic cell transplantation. *Transplant Cell Ther* 2022; 28:846e1–846e9.
9. Dekker IM, Bruggink H, van der Meij BS, Wierdsma NJ. State of the art: the role of citrulline as biomarker in patients with chemotherapy- or graft-versus-host-disease-induced mucositis. *Curr Opin Clin Nutr Metab Care* 2021; 24:416–427.

This update focuses on the results of studies, which have used plasma citrulline level as a sensitive biomarker of functional enterocyte mass in patients with oncological disease receiving high-dose chemotherapy or undergoing allo-HCT.

10. van der Velden WJ, Herbers AH, Bruggemann RJ, *et al.* Citrulline and albumin as biomarkers for gastrointestinal mucositis in recipients of hematopoietic SCT. *Bone Marrow Transplant* 2013; 48:977–981.
11. Hueso T, Coiteux V, Joncquel Chevalier Curt M, *et al.* Citrulline and monocyte-derived macrophage reactivity before conditioning predict acute graft-versus-host disease. *Biol Blood Marrow Transplant* 2017; 23:913–921.
12. Hueso T, Gauthier J, Joncquel Chevalier-Curt M, *et al.* Association between low plasma level of citrulline before allogeneic hematopoietic cell transplantation and severe gastrointestinal graft vs host disease. *Clin Gastroenterol Hepatol* 2018; 16:908–917; e902.
13. Hueso T, Ekpe K, Mayeur C, *et al.* Impact and consequences of intensive chemotherapy on intestinal barrier and microbiota in acute myeloid leukemia: the role of mucosal strengthening. *Gut Microbes* 2020; 12:1800897.
14. Rashidi A, Ebadi M, Rehman TU, *et al.* Lasting shift in the gut microbiota in patients with acute myeloid leukemia. *Blood Adv* 2022; 6:3451–3457.
15. Tyszk M, Maciejewska-Markiewicz D, Bilinski J, *et al.* Increased intestinal permeability and stool zonulin, calprotectin and beta-defensin-2 concentrations in allogeneic hematopoietic cell transplantation recipients. *Int J Mol Sci* 2022; 23:15962.
16. Weber D, Weber M, Meedt E, *et al.* Reg3alpha concentrations at day of allogeneic stem cell transplantation predict outcome and correlate with early antibiotic use. *Blood Adv* 2023; 7:1326–1335.
17. Peled JU, Gomes ALC, Devlin SM, *et al.* Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med* 2020; 382:822–834.
18. Yamamoto A, Kambara Y, Fujiwara H. Impact of oral microbiota on pathophysiology of GVHD. *Front Immunol* 2023; 14:1132983.
19. Ingham AC, Kielsen K, Mordhorst H, *et al.* Microbiota long-term dynamics and prediction of acute graft-versus-host disease in pediatric allogeneic stem cell transplantation. *Microbiome* 2021; 9:148.
20. Heidrich V, Bruno JS, Knebel FH, *et al.* Dental biofilm microbiota dysbiosis is associated with the risk of acute graft-versus-host disease after allogeneic hematopoietic stem cell transplantation. *Front Immunol* 2021; 12:692225.
21. Szovati S, Morrison CF, Couch SC. Nutritional status of allogeneic hematopoietic stem cell transplant recipients and posttransplant outcomes. *Nutr Cancer* 2023; 75:1200–1210.
22. Trifilio S, Helenowski I, Giel M, *et al.* Questioning the role of a neutropenic diet following hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant* 2012; 18:1385–1390.
23. Ma Y, Lu X, Liu H. Neutropenic diet cannot reduce the risk of infection and mortality in oncology patients with neutropenia. *Front Oncol* 2022; 12:836371.
24. Toenges R, Greinix H, Lawitschka A, *et al.* Current practice in nutrition after allogeneic hematopoietic stem cell transplantation: results from a survey among hematopoietic stem cell transplant centers. *Clin Nutr* 2021; 40:1571–1577.
25. Muratore E, Leardini D, Baccelli F, *et al.* Nutritional modulation of the gut microbiome in allogeneic hematopoietic stem cell transplantation recipients. *Front Nutr* 2022; 9:993668.
26. Qiao X, Bilinski J, Wang L, *et al.* Safety and efficacy of fecal microbiota transplantation in the treatment of graft-versus-host disease. *Bone Marrow Transplant* 2023; 58:10–19.
27. Holmes ZC, Tang H, Liu C, *et al.* Prebiotic galactooligosaccharides interact with mouse gut microbiota to attenuate acute graft-versus-host disease. *Blood* 2022; 140:2300–2304.
28. Docampo MD, da Silva MB, Lazrak A, *et al.* Alloreactive T cells deficient of the short-chain fatty acid receptor GPR109A induce less graft-versus-host disease. *Blood* 2022; 139:2392–2405.
29. Stein-Thoeringer CK, Nichols KB, Lazrak A, *et al.* Lactose drives Enterococcus expansion to promote graft-versus-host disease. *Science* 2019; 366:1143–1149.
30. Pan P, Atkinson SN, Taylor B, *et al.* Retinoic acid signaling modulates recipient gut barrier integrity and microbiota after allogeneic hematopoietic stem cell transplantation in mice. *Front Immunol* 2021; 12:749002.
31. Daloglu H, Uygun V, Ozturkmen S, *et al.* Pretransplantation vitamin D deficiency increases acute graft-versus-host disease after hematopoietic stem cell transplantation in thalassemia major patients. *Clin Transplant* 2023; 37:e14874.

In this retrospective study including 100 pediatric patients, low vitamin D levels before allo-HCT was an independent risk factor of grade II–IV GvHD.

32. Bhandari R, Aguayo-Hirald P, Malvar J, *et al.* Ultra-high dose vitamin D in pediatric hematopoietic stem cell transplantation: a nonrandomized controlled trial. *Transplant Cell Ther* 2021; 27:1001e1–1001e9.
33. Macedo R, Pasin C, Ganetsky A, *et al.* Vitamin D deficiency after allogeneic hematopoietic cell transplantation promotes T-cell activation and is inversely associated with an EZH2-ID3 signature. *Transplant Cell Ther* 2022; 28:18.e11–18.e10.
34. Matos C, Peter K, Weich L, *et al.* Anti-thymocyte globulin treatment augments 1,25-dihydroxyvitamin D3 serum levels in patients undergoing hematopoietic stem cell transplantation. *Front Immunol* 2021; 12:803726.

35. Matos C, Mamilos A, Shah PN, *et al.* Downregulation of the vitamin D receptor expression during acute gastrointestinal graft versus host disease is associated with poor outcome after allogeneic stem cell transplantation. *Front Immunol* 2022; 13:1028850.
 36. Gjaerde LK, Ostrowski SR, Jorgensen NR, *et al.* Pretransplantation vitamin E levels and acute graft-versus-host disease after nonmyeloablative allogeneic hematopoietic cell transplantation. *Transpl Immunol* 2022; 74:101650.
 37. Gonzales F, Bruno B, Alarcon Fuentes M, *et al.* Better early outcome with enteral rather than parenteral nutrition in children undergoing MAC allo-SCT. *Clin Nutr* 2018; 37:2113–2121.
 38. Seguy D, Duhamel A, Rejeb MB, *et al.* Better outcome of patients undergoing enteral tube feeding after myeloablative conditioning for allogeneic stem cell transplantation. *Transplantation* 2012; 94:287–294.
 39. Zama D, Gori D, Muratore E, *et al.* Enteral versus parenteral nutrition as nutritional support after allogeneic hematopoietic stem cell transplantation: a systematic review and meta-analysis. *Transplant Cell Ther* 2021; 27:180e1–180e8.
- This meta-analysis based on eight clinical studies shows that enteral nutrition reduces the incidence GvHD as well as gastrointestinal GvHD compared with parenteral nutrition.
40. Iyama S, Tatsumi H, Shiraishi T, *et al.* Possible clinical outcomes using early enteral nutrition in individuals with allogeneic hematopoietic stem cell transplantation: a single-center retrospective study. *Nutrition* 2021; 83:111093.
 41. Morello E, Guarinoni MG, Arena F, *et al.* A systematic review of the literature and perspectives on the role of biomarkers in the management of malnutrition after allogeneic hematopoietic stem cell transplantation. *Front Immunol* 2020; 11:535890.
 42. Evans J, Green D, Connor GO, *et al.* Nutritional support practices and opinions toward gastrostomy use in pediatric bone marrow transplant centers: a national survey. *Nutrition* 2022; 95:111556.
 43. Kairiene I, Vaisvilas M, Vasciunaite A, *et al.* Impact of percutaneous endoscopic gastrostomy on pediatric bone marrow transplantation outcomes: retrospective single-center cohort study. *JPEN J Parenter Enteral Nutr* 2023; 47:390–398.
 44. Alsalamah S, Alramyan R, Alakel R, *et al.* The outcome and complications of total parenteral nutrition in pediatric hematopoietic stem cell transplantation. *Pediatr Transplant* 2022; 26:e14198.
 45. Khodr J, Zerbib P, Rogosnitzky M, *et al.* Diverting enterostomy improves overall survival of patients with severe steroid-refractory gastrointestinal acute graft-versus-host disease. *Ann Surg* 2021; 274:773–779.
- In this comparative retrospective study, diverting enterostomy significantly increased 1-year and 2-year overall survival compared with patients treated with standard of care in case of steroid-refractory gastrointestinal GvHD (54 versus 5% and 31 versus 2.5%, respectively).